

Sales Performance Analysis — Task 2 Submission

Alfido Tech Data Analytics Internship Program

Code & Files (GitHub): <https://github.com/ManjuVikram/sales-performance-analysis>

Executive Summary

Analyzed 9,800 sales line items across 4,922 orders from 793 customers (Jan 2015–Dec 2018) for Alfido Tech, covering \$2.26M in total sales. Sales grew overall: after a slight dip in 2016 (–4.3%), growth accelerated in 2017 (+30.6%) and continued in 2018 (+20.3%). Strong seasonality shows September, November, and December as peak months. Technology drives the most revenue per order while Office Supplies drives the most order volume. Revenue is broadly distributed across customers (~50% of customers reach 80% of revenue) — a healthier spread than typical retail concentration. Shipping is dominated by the slowest option (Standard Class, 59.8% of orders).

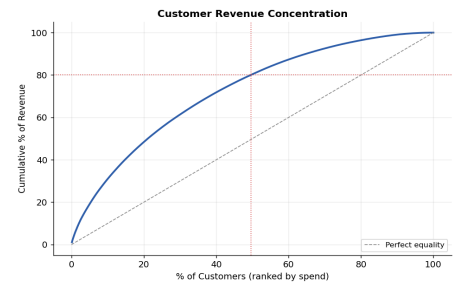
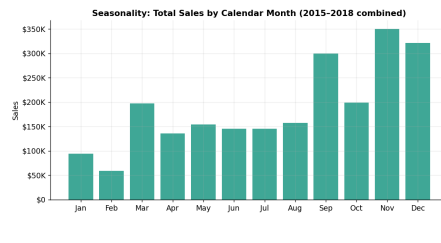
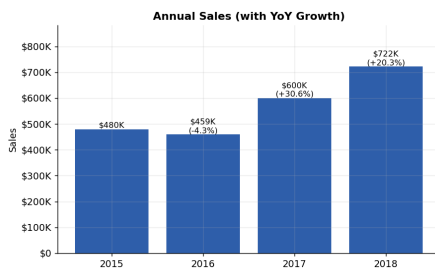
Key Findings

- Sales grew overall: 2016 dip (–4.3%) then +30.6% (2017), +20.3% (2018).
- Strong seasonality — Sep/Nov/Dec are peak months, 2–3x mid-year.
- Technology: highest revenue/order. Office Supplies: highest order volume.
- Revenue broadly spread — ~50% of customers drive 80% of revenue.
- Standard Class dominates (60% of orders) despite slowest delivery (~5 days).

Top Recommendations

- Plan inventory/staffing around the Sep/Nov/Dec seasonal peak.
- Promote faster shipping tiers more assertively at checkout.
- Double down on Technology as a margin/value driver via bundling.
- Use Office Supplies as a traffic driver, then cross-sell higher-value items.
- Investigate and target underperforming states for incremental revenue.

Key Visualizations



Annual sales & YoY growth | Seasonality by month | Customer revenue concentration

Notebook Screenshots

Sales Performance Analysis — Alfido Tech

Task 2 of 3 - Internship Data Analytics Program

Dataset: Superstore Sales Dataset (Kaggle) - [superstore_final_dataset.csv](#)

Goal: Analyze sales transactions to understand revenue trends, top-performing categories/products/regions, seasonality, and shipping performance, and provide actionable recommendations for Alfido Tech.

Contents

- Setup & Data Loading
- Data Cleaning
- Summary Statistics & Data Types
- Feature Engineering (date parts, shipping delay)
- Sales Trend Analysis (yearly, monthly, seasonality)
- Category, Sub-Category & Product Performance
- Regional & State Performance
- Customer Segment & Concentration Analysis
- Shipping Mode Analysis
- Key Findings & 5 Actionable Recommendations

1. Setup & Data Loading

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import warnings
warnings.filterwarnings('ignore')

# Import the dataset
superstore = pd.read_csv('superstore_final_dataset.csv')

# Display the first few rows
superstore.head()
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country
0	1	CA-2017-132356	8/11/2017	11/11/2017	CG-12520	Claire Guitre	Consumer	United States
1	2	CA-2017-132356	8/11/2017	11/11/2017	CG-12520	Claire Guitre	Consumer	United States
2	3	CA-2017-136688	12/6/2017	16/06/2017	DV-11045	Darvin Van Hout	Corporate	United States
3	4	US-2016-108668	11/10/2016	18/10/2016	SO-20335	Sean O'Donnell	Consumer	United States
4	5	US-2016-108668	11/10/2016	18/10/2016	SO-20335	Sean O'Donnell	Consumer	United States

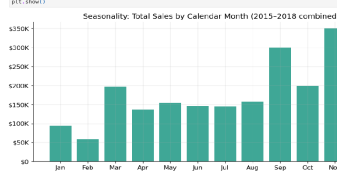
```
-----
Year 2015 2016 2017 2018
Sales 480 459 600 722
YoY % change:
Order Year
2015 480
2016 459
2017 600
2018 722
Name: Sales, dtype: float64
```

Annual Sales (with YoY Growth)



Finding: sales dipped slightly in 2016 (–4.3%) but then grew strongly in 2017 (+30.6%) and 2018 (+20.3%) — a healthy, accelerating growth trajectory overall.

```
In [11]: month_sales = df.groupby('Order_Month')['Sales'].sum()
fig, ax = plt.subplots(figsize=(10, 10))
ax.bar(month_sales.index, month_sales.values, color='#F08080')
ax.set_xlabel('Month')
ax.set_ylabel('Sales ($K)')
ax.set_title('Seasonality: Total Sales by Calendar Month (2015-2018 combined)')
plt.gcf().set_fig_label('Seasonality: Total Sales by Calendar Month (2015-2018 combined)')
plt.show()
```



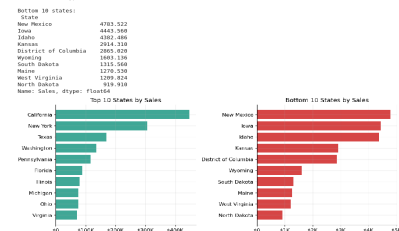
Finding: clear seasonality — September (back-to-school), and November/December (holiday season) are by far the strongest months, each roughly 2–3x a typical mid-year month.

6. Category, Sub-Category & Product Performance

```
In [12]: cat = df.groupby('Category')['Sales'].agg(['sum', 'count']).sort_values('sum', ascending=False)
```

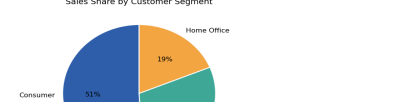
```
plt.figure(figsize=(10, 10))
plt.show()

Top 10 states:
State  Sales
California 440081.4875
New York 386081.1470
Texas 386081.1470
Washington 33206.0000
Pennsylvania 33206.0000
Florida 29206.0000
Illinois 29206.0000
Michigan 29206.0000
Ohio 29206.0000
Virginia 29206.0000
Name: Sales, dtype: float64
```



8. Customer Segment & Concentration Analysis

```
In [17]: segment = df.groupby('Segment')['Sales'].agg(['sum', 'count']).sort_values('sum', ascending=False)
fig, ax = plt.subplots(figsize=(10, 10))
ax.bar(segment.index, segment['sum'], color='#F08080')
ax.set_xlabel('Segment')
ax.set_ylabel('Sales ($K)')
ax.set_title('Customer Segment & Concentration Analysis')
plt.gcf().set_fig_label('Customer Segment & Concentration Analysis')
plt.show()
```



Data loading & overview | Annual/seasonal trend analysis | State & segment breakdown

Full notebook, executed outputs, and the detailed 10-page analytical report are available in the GitHub repo above.